Notes on Singular Value Decomposition

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Abstract

We collect a few elementary facts about the singular value decomposition (SVD) of matrices. In particular, we present three approaches used by different authors in the history to establish the existence of SVD.

Keywords. Singular value decomposition, Eigenvalue decomposition, Deflation, Jordan-Wielandt matrix

Notation. Throughout this document, m, n, k, i, and j are positive integers; r is a nonnegative integer. We use $\|\cdot\|$ to denote the 2-norm for vectors and matrices. In inline equations, the MATLAB-style notation $[a;b]$ stands for a vertical array with a and b being its entries. The identity matrix is represented by I, and it may take a subscript to indicate its order when necessary. Given any matrix A, we use $A_{i,j}$ to signify its (i,j) entry.

1 Eigenvalue decomposition

The following theorem and definition can be found in textbooks on linear algebra.

Theorem 1. *For any Hermitian matrix $A \in \mathbb{C}^{n\times n}$, there exists a unitary matrix $U \in \mathbb{C}^{n\times n}$ and a diagonal matrix $\Lambda \in \mathbb{R}^{n\times n}$ such that*

$$A \;=\; U\Lambda U^{\mathsf{H}}.$$

If A is real, then we can require that U is real. Indeed, $\Lambda_{1,1}, \ldots, \Lambda_{n,n}$ are the eigenvalues of A, multiplicity included, and the j-th column of U is an eigenvector of A associated with $\Lambda_{j,j}$.

Definition 1. Suppose that $A \in \mathbb{C}^{n\times n}$ is an Hermitian matrix and $A = U\Lambda U^{\mathsf{H}}$.

1. $U\Lambda U^{\mathsf{H}}$ is called an eigenvalue decomposition of A, provided that $U \in \mathbb{C}^{n\times n}$ is a unitary matrix, and $\Lambda \in \mathbb{R}^{n\times n}$ is a diagonal matrix.

2. $U\Lambda U^{\mathsf{H}}$ is called a compact eigenvalue decomposition of A, provided that $U \in \mathbb{C}^{n\times r}$ is a matrix with $U^{\mathsf{H}}U = I_r$, and $\Lambda \in \mathbb{R}^{r\times r}$ is a diagonal matrix whose diagonal entries are nonzero.

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2 Singular value decomposition (SVD)

Definition 2. Suppose that $A \in \mathbb{C}^{m\times n}$ is a matrix with $\operatorname{rank}(A) = r$, and $A = U\Sigma V^{\mathsf{H}}$.

1. $U\Sigma V^{\mathsf{H}}$ is called an singular value decomposition of A, provided that $U \in \mathbb{C}^{m\times m}$ and $V \in \mathbb{C}^{n\times n}$ are unitary matrices, and $\Sigma \in \mathbb{R}^{m\times n}$ is a matrix whose first r diagonal entries (if $r \geq 1$) are positive while all the other entries are zero.

2. When $r \geq 1$, $U\Sigma V^{\mathsf{H}}$ is called a compact singular value decomposition of A, provided that $U \in \mathbb{C}^{m\times r}$ and $V \in \mathbb{C}^{n\times r}$ satisfy $U^{\mathsf{H}}U = V^{\mathsf{H}}V = I_r$, and $\Sigma \in \mathbb{R}^{r\times r}$ is a diagonal matrix whose diagonal entries are positive.

3. When $m \geq n$, $U\Sigma V^{\mathsf{H}}$ is called a thin (or economy-sized) singular value decomposition of A, provided that $U \in \mathbb{C}^{m\times n}$ and $V \in \mathbb{C}^{n\times n}$ satisfy $U^{\mathsf{H}}U = V^{\mathsf{H}}V = I_n$, and $\Sigma \in \mathbb{R}^{m\times m}$ is a matrix whose first r diagonal entries (if $r \geq 1$) are positive while all the other entries are zero. When $n \geq m$, a thin singular value decomposition can be defined similarly.

Remark 1. *Let $U\Sigma V^{\mathsf{H}}$ be an SVD of $A \in \mathbb{R}^{m\times n}$ and $\sigma_i = \Sigma_{i,i}$, where $1 \leq i \leq \min\{m, n\}$. Then $\sigma_1, \ldots, \sigma_r$ are called the (nonzero) singular values of A. It is often convenient to regard $\sigma_{r+1} = \cdots = \sigma_{\min\{m,n\}} = 0$ also as singular values of A.*

Remark 2. *If $U\Sigma V^{\mathsf{H}}$ is a (compact) SVD of A, then $AV = U\Sigma$ and $A^{\mathsf{H}}U = V\Sigma$. Let $\sigma_i = \Sigma_{i,i}$, u_i be the i-th column of U, and v_i be the i-th column of V. Then $Av_i = \sigma_i u_i$ and $A^{\mathsf{H}}u_i = \sigma_i v_i$; u_i and v_i are called a pair of left and right singular vectors of A associated with the singular value σ_i.*

Remark 3. *If $U\Sigma V^{\mathsf{H}}$ is a compact SVD of $A \in \mathbb{C}^{m\times n}$, then we can extend it to an SVD*

$$(U\ \tilde{U}) \begin{pmatrix} \Sigma & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} V^{\mathsf{H}} \\ \tilde{V}^{\mathsf{H}} \end{pmatrix},$$

where $\tilde{U} \in \mathbb{C}^{m\times(m-r)}$ is any matrix such that $(U\ \tilde{U})$ is unitary, and $\tilde{V} \in \mathbb{C}^{n\times(n-r)}$ is any matrix such that $(V\ \tilde{V})$ is unitary. Conversely, if $U\Sigma V^{\mathsf{H}}$ is an SVD of $A \neq 0$, we can obtain a compact SVD by dropping the zero diagonal entries of Σ and the corresponding columns of U and V.

Remark 4. *The names* thin *SVD,* economy-sized *SVD, and* compact *SVD are adopted from [1, Section 6.1]. Other names exist. In [11, Lecture 4], a thin SVD is called a* reduced *SVD instead. In [5, Theorem 7.3.2.], "thin SVD" means what we call "compact SVD" here.*

2.1 Uniqueness of (compact) SVD

Theorem 2. *Suppose that $U\Sigma V^{\mathsf{H}}$ is a compact SVD of a matrix A.*

1. *$U\Sigma^2 U^{\mathsf{H}}$ is a compact eigenvalue decomposition of AA^{H}, and hence the diagonal entries of Σ^2 are the positive eigenvalues of AA^{H}, multiplicity included.*

2. *$V\Sigma^2 V^{\mathsf{H}}$ is a compact eigenvalue decomposition of $A^{\mathsf{H}}A$, and hence the diagonal entries of Σ^2 are the positive eigenvalues of $A^{\mathsf{H}}A$, multiplicity included.*

Lemma 1. *Consider matrices $A \in \mathbb{C}^{n\times n}$ and $\Lambda \in \mathbb{C}^{n\times n}$ with Λ being diagonal.*

1. *$\Lambda A = A\Lambda$ if and only if $A_{i,j} = 0$ for any i and j such that $\Lambda_{i,i} \neq \Lambda_{j,j}$.*

2. *If Λ is nonnegative and $\Lambda A = A\Lambda$, then $\Lambda^p A = A\Lambda^p$ for any $p \geq 0$.*

Proof. Because $\Lambda A = A\Lambda$ if and only if $\Lambda_{i,i}A_{i,j} = A_{i,j}\Lambda_{j,j}$ for any $i, j \in \{1, \ldots, n\}$. □

Lemma 2. *Let U_1, $U_2 \in \mathbb{C}^{n\times r}$ satisfy $U_1^{\mathsf{H}}U_1 = U_2^{\mathsf{H}}U_2 = I_r$ and* $\mathrm{range}(U_1) = \mathrm{range}(U_2)$.

1. *$U_1U_1^{\mathsf{H}} = U_2U_2^{\mathsf{H}}$, both being the orthogonal projection onto* $\mathrm{range}(U_1) = \mathrm{range}(U_2)$.

2. *$W = U_1^{\mathsf{H}}U_2$ is a unitary matrix and $U_1W = U_2$.*

Proof. Since $U_1^{\mathsf{H}}U_1 = I_r$, $U_1U_1^{\mathsf{H}}$ is the orthogonal projection onto $\mathrm{range}(U_1) = \mathrm{range}(U_2)$ (see, e.g., [4]). In addition, $U_1W = U_1U_1^{\mathsf{H}}U_2 = U_2$, and $W^{\mathsf{H}}W = U_2^{\mathsf{H}}U_1W = U_2^{\mathsf{H}}U_2 = I_r$. □

Theorem 3. *Let $U_i \in \mathbb{C}^{m\times r}$ and $V_i \in \mathbb{C}^{n\times r}$ satisfy $U_i^{\mathsf{H}}U_i = V_i^{\mathsf{H}}V_i = I_r$ $(i = 1, 2)$, and $\Sigma \in \mathbb{C}^{r\times r}$ be a diagonal matrix whose diagonal entries are positive. Then $U_1\Sigma V_1^{\mathsf{H}} = U_2\Sigma V_2^{\mathsf{H}}$ if and only if there exists a unitary matrix $W \in \mathbb{C}^{r\times r}$ such that $U_2 = U_1W$, $V_2 = V_1W$, and $\Sigma W = W\Sigma$.*

Proof. We will focus on the "only if" part since the "if" part is trivial. Assuming that $U_1\Sigma V_1^{\mathsf{H}} = U_2\Sigma V_2^{\mathsf{H}}$, we will show that $W = U_1^{\mathsf{H}}U_2 \in \mathbb{C}^{r\times r}$ fulfills all the desired requirements. Observe that both ΣV_1^{H} and ΣV_2^{H} have full column rank. Hence

$$\mathrm{range}(U_1) \;=\; \mathrm{range}(U_1\Sigma V_1^{\mathsf{H}}) \;=\; \mathrm{range}(U_2\Sigma V_2^{\mathsf{H}}) \;=\; \mathrm{range}(U_2).$$

By Lemma 2, W is a unitary matrix and $U_1W = U_2$. It remains to show that $\Sigma W = W\Sigma$ and $V_2 = WV_1$. Recalling that $V_1^{\mathsf{H}}V_1 = V_2^{\mathsf{H}}V_2 = I_r$, we have

$$U_1\Sigma^2U_1^{\mathsf{H}} \;=\; (U_1\Sigma V_1^{\mathsf{H}})(U_1\Sigma V_1^{\mathsf{H}})^{\mathsf{H}} \;=\; (U_2\Sigma V_2^{\mathsf{H}})(U_2\Sigma V_2^{\mathsf{H}})^{\mathsf{H}} \;=\; U_2\Sigma^2U_2^{\mathsf{H}}.$$

Hence

$$\Sigma^2W \;=\; \Sigma^2U_1^{\mathsf{H}}U_2 \;=\; U_1^{\mathsf{H}}(U_1\Sigma^2U_1^{\mathsf{H}})U_2 \;=\; U_1^{\mathsf{H}}(U_2\Sigma^2U_2^{\mathsf{H}})U_2 \;=\; U_1^{\mathsf{H}}U_2\Sigma^2 \;=\; W\Sigma^2$$

Thus $\Sigma W = W\Sigma$ by Lemma 1. Finally, since $V_1\Sigma U_1^{\mathsf{H}} = (U_1\Sigma V_1^{\mathsf{H}})^{\mathsf{H}} = (U_2\Sigma V_2^{\mathsf{H}})^{\mathsf{H}} = V_2\Sigma U_2^{\mathsf{H}}$,

$$V_2 \;=\; (V_2\Sigma U_2^{\mathsf{H}})(U_2\Sigma^{-1}) \;=\; (V_1\Sigma U_1^{\mathsf{H}})(U_2\Sigma^{-1}) \;=\; V_1\Sigma W\Sigma^{-1} \;=\; V_1W\Sigma\Sigma^{-1} \;=\; V_1W.$$

The proof is complete. □

2.2 Existence of SVD

Here we present three independent ways of establishing the existence of SVD (see Theorems 4, 5, 6). They were used by different authors in the history [10]. Among these three approaches, Jordan's method does not depend on eigenvalue decomposition, but the other two do.

Note that it suffices to prove the existence of the compact SVD, from which an SVD can be constructed easily.

2.2.1 Jordan's deflation approach [6]

Lemma 3. *Given a nonzero matrix $A \in \mathbb{C}^{m\times n}$, let $(u, v) \in \mathbb{C}^m \times \mathbb{C}^n$ be a solution of*

$$\max\,\{\Re(x^{\mathsf{H}}Ay) : \|x\| = \|y\| = 1,\ x \in \mathbb{C}^m,\ y \in \mathbb{C}^n\},$$

and $\sigma = \Re(u^{\mathsf{H}}Av)$. Then $Av = \sigma u$, $A^{\mathsf{H}}u = \sigma v$, and $\sigma > 0$.

Proof. Since $A \neq 0$, it is obvious that $\sigma > 0$. Hence $Av \neq 0$. According to the definition of u,

$$\Re(u^{\mathsf{H}}Av) \geq \Re((Av/\|Av\|)^{\mathsf{H}}\,Av) \;=\; \|Av\| \;=\; \|u\|\|Av\|.$$

By the Cauchy-Schwarz inequality, there exists a scalar $\lambda > 0$ such that $\lambda u = Av$. Hence

$$\sigma \;=\; \Re(u^{\mathsf{H}}Av) \;=\; \Re(\lambda\|u\|^2) \;=\; \lambda.$$

Thus $Av = \sigma u$. Similarly, we can prove $A^{\mathsf{H}}u = \sigma v$ using the fact that

$$\Re(u^{\mathsf{H}}Av) \;\geq\; \Re(u^{\mathsf{H}}A\,(A^{\mathsf{H}}u/\|A^{\mathsf{H}}u\|)) \;=\; \|A^{\mathsf{H}}u\| = \|A^{\mathsf{H}}u\|\|v\|.$$

□

Remark 5. *When Lemma 3 is applied in the proof of Theorem 4 later, we only need the existence of unit vectors $u \in \mathbb{C}^m$, $v \in \mathbb{C}^n$, and a scalar $\sigma > 0$ such that $Av = \sigma u$ and $A^{\mathsf{H}}u = \sigma v$. The existence can be established in other ways. Here are two examples.*

1. *Let $\sigma = (\lambda_{\max}(AA^{\mathsf{H}}))^{\frac{1}{2}}$, $u \in \mathbb{C}^{m\times m}$ be an eigenvector of AA^{H} associated with $\lambda_{\max}(AA^{\mathsf{H}})$, and $v = A^{\mathsf{H}}u/\sigma$. Then $Av = AA^{\mathsf{H}}u/\sigma = \sigma^2 u/\sigma = \sigma u$, and $A^{\mathsf{H}}u = \sigma v$. This is the approach used in the proofs of [11, Theorem 4.1] and [7, Theorem 1].*

2. *Let $\sigma = \lambda_{\max}(J)$ with J being the Jordan-Wielandt form of A (see (2.6)), $w \in \mathbb{C}^{m+n}$ be an eigenvector associated with σ, $x \in \mathbb{C}^m$ consist of the first m entries of w, and $y \in \mathbb{C}^n$ consist of the last n. Then we can verify that $Ay = \sigma x$ and $A^{\mathsf{H}}x = \sigma y$. Meanwhile, $A(-y) = -\sigma x$, and $A^{\mathsf{H}}x = -\sigma(-y)$, making $[x; -y]$ an eigenvector of J associated with $-\sigma \neq \sigma$. Since J is Hermitian, we know that w and $[x; -y]$ are orthogonal, and hence $x^{\mathsf{H}}x - y^{\mathsf{H}}y = 0$. Thus $\|x\| = \|y\|$, which are nonzero since $w \neq 0$. Finally, let $u = x/\|x\|$ and $v = y/\|y\|$.*

Theorem 4. *Any $A \in \mathbb{C}^{m\times n}$ has an SVD $U\Sigma V^{\mathsf{H}}$ as defined in Definition 2.*

Proof. Assume without loss of generality that $A \neq 0$. We prove by an induction on $\min\{m, n\}$.

1. If $\min\{m, n\} = 1$, then A is either a row or a column. If A is a column, let U be a unitary matrix whose first column is $A/\|A\|$, $\Sigma = e_1$ (i.e., the first canonical coordinate vector), and $V = \|A\|$. Then $U\Sigma V^{\mathsf{H}}$ is an SVD of A. If A is a row, the decomposition can be found similarly.

2. Assume that the conclusion holds when $\min\{m, n\} = k$. Let us consider the scenario where $\min\{m, n\} = k + 1$. Let A be a matrix in $\mathbb{C}^{m\times n}$. By Lemma 3, there exist unit vectors $u \in \mathbb{C}^m$, $v \in \mathbb{C}^n$, and a scalar $\sigma > 0$ such that

$$Av \;=\; \sigma u, \quad A^{\mathsf{H}}u \;=\; \sigma v. \tag{2.1}$$

Let $U \in \mathbb{C}^{m\times m}$ be a unitary matrix whose first column is u, and $V \in \mathbb{C}^{n\times n}$ be a unitary matrix whose first column is v. It is then straightforward to check that

$$U^{\mathsf{H}}AV \;=\; \begin{pmatrix} \sigma & 0 \\ 0 & \hat{A} \end{pmatrix}, \tag{2.2}$$

where $\hat{A}$ is a matrix in $\mathbb{C}^{(m-1)\times(n-1)}$. If $\hat{A} = 0$, then (2.2) provides an SVD of A. Otherwise, since $\min\{m-1, n-1\} = \min\{m, n\} - 1$, we know from the induction hypothesis that $\hat{A}$ has an SVD $\hat{U}\hat{\Sigma}\hat{V}^{\mathsf{H}}$. Consequently,

$$A \;=\; U(U^{\mathsf{H}}AV)V^{\mathsf{H}} \;=\; U\begin{pmatrix} \sigma & 0 \\ 0 & \hat{U}\hat{\Sigma}\hat{V}^{\mathsf{H}} \end{pmatrix}V^{\mathsf{H}} \;=\; \left[U\begin{pmatrix} 1 & 0 \\ 0 & \hat{U} \end{pmatrix}\right]\begin{pmatrix} \sigma & 0 \\ 0 & \hat{\Sigma} \end{pmatrix}\left[\begin{pmatrix} 1 & 0 \\ 0 & \hat{V} \end{pmatrix}^{\mathsf{H}} V^{\mathsf{H}}\right]. \tag{2.3}$$

It is easy to verify that the right-hand side of (2.3) provides an SVD of A. This completes the induction. □

Remark 6. *We can also prove Theorem 4 by an induction on* $\operatorname{rank}(A)$*. When* $\operatorname{rank}(A) = 0$*, the desired conclusion is trivial. Assume that the conclusion holds when* $\operatorname{rank}(A) \le k$*. Let us consider the scenario with* $\operatorname{rank}(A) = k+1$*. By Lemma 3, there exists unit vectors* $u \in \mathbb{C}^m$*,* $v \in \mathbb{C}^n$*, and a scalar* $\sigma > 0$ *fulfilling* (2.1)*. Define* $\hat{A} = A - \sigma uv^{\mathsf{H}}$*. Then it is easy to check that* $\ker(A) \subset \ker(\hat{A})$ *and* $v \in \ker(\hat{A})$*. Since* $v \in \operatorname{range}(A^{\mathsf{H}}) \perp \ker(A)$*, we know that* $\dim\ker(\hat{A}) \ge \dim\ker(A) + 1$*. Thus* $\operatorname{rank}(\hat{A}) \le \operatorname{rank}(A) - 1$*. If* $\hat{A} = 0$*, we are done. Otherwise, by the induction hypothesis,* $\hat{A}$ *has a compact SVD* $\hat{U}\hat{\Sigma}\hat{V}^{\mathsf{H}}$*. Consequently,*

$$A \;=\; \sigma uv^{\mathsf{H}} + \hat{A} \;=\; \sigma uv^{\mathsf{H}} + \hat{U}\hat{\Sigma}\hat{V}^{\mathsf{H}} \;=\; (u\ \hat{U})\begin{pmatrix} \sigma & 0 \\ 0 & \hat{\Sigma} \end{pmatrix}(v\ \hat{V})^{\mathsf{H}}. \tag{2.4}$$

Noting that $\hat{A}v = 0$*,* $\hat{A}^{\mathsf{H}}u = 0$*, and* $\hat{\Sigma}$ *is nonsingular, we can see that* $\hat{V}^{\mathsf{H}}v = 0$ *and* $\hat{U}^{\mathsf{H}}u = 0$*. Thus the columns of* $(u\ \hat{U})$ *are orthonormal, and so are those of* $(v\ \hat{V})$*. Hence* (2.4) *provides a compact SVD for* A*, which can be extended to an SVD. The induction is complete.*

2.2.2 The Eckart-Young approach [2]

Lemma 4. *Suppose that* $A \in \mathbb{C}^{m\times n}$ *and* $B \in \mathbb{C}^{n\times k}$*.*

1. $\|AB\|_{\mathrm{F}} \le \|A\|\|B\|_{\mathrm{F}}$*, and the equality holds if and only if* $A^{\mathsf{H}}AB = \|A\|^2 B$*.*
2. $\|AB\|_{\mathrm{F}} \le \|A\|_{\mathrm{F}}\|B\|$*, and the equality holds if and only if* $ABB^{\mathsf{H}} = \|B\|^2 A$*.*

Remark 7. *Recall that* $\|\cdot\|$ *denotes the* 2*-norm for matrices.*

Proof. Let $C \in \mathbb{C}^{n\times n} = (\|A\|^2 I - A^{\mathsf{H}}A)^{\frac{1}{2}}$, which is well defined since $\|A\|^2 I - A^{\mathsf{H}}A$ is positive semidefinite. Then

$$\|A\|^2\|B\|_{\mathrm{F}}^2 - \|AB\|_{\mathrm{F}}^2 \;=\; \operatorname{tr}(\|A\|^2 B^{\mathsf{H}}B) - \operatorname{tr}(B^{\mathsf{H}}A^{\mathsf{H}}AB) \;=\; \operatorname{tr}(B^{\mathsf{H}}C^2B) \;\ge\; 0.$$

Thus $\|AB\|_{\mathrm{F}} \le \|A\|\|B\|_{\mathrm{F}}$, and

$$\|AB\|_{\mathrm{F}} = \|A\|\|B\|_{\mathrm{F}} \iff B^{\mathsf{H}}C^2B = 0 \iff C^2B = 0 \iff A^{\mathsf{H}}AB = \|A\|^2 B.$$

The proof concerning $\|AB\|_{\mathrm{F}} \le \|A\|_{\mathrm{F}}\|B\|$ is similar. □

Lemma 5. *Let A, B, U, and V be complex matrices such that both $U^{\mathsf{H}}AV$ and UBV^{H} are well defined. If $\|A\|_{\mathrm{F}} = \|B\|_{\mathrm{F}}$ and $\|U\|\|V\| = 1$, then $A = UBV^{\mathsf{H}}$ if and only if $B = U^{\mathsf{H}}AV$.*

Proof. Without loss of generality, we suppose that $\|U\| = \|V\| = 1$. Otherwise, consider $U/\|U\|$ and $V/\|V\|$ instead of U and V respectively.

Assume that $A = UBV^{\mathsf{H}}$. Since $\|U\| = \|V\| = 1$ and $\|A\|_{\mathrm{F}} = \|B\|_{\mathrm{F}}$, we have

$$\min\{\|UB\|_{\mathrm{F}},\ \|BV^{\mathsf{H}}\|_{\mathrm{F}}\} \ \geq\ \|UBV^{\mathsf{H}}\|_{\mathrm{F}} \ =\ \|A\|_{\mathrm{F}} = \|B\|_{\mathrm{F}}.$$

Hence Lemma 4 ensures

$$U^{\mathsf{H}}UB \ =\ B, \quad BV^{\mathsf{H}}V \ =\ B.$$

Therefore,

$$U^{\mathsf{H}}AV \ =\ U^{\mathsf{H}}UBV^{\mathsf{H}}V \ =\ BV^{\mathsf{H}}V \ =\ B.$$

In the same way, $B = U^{\mathsf{H}}AV$ implies $A = UBV^{\mathsf{H}}$. □

Theorem 5. *Let $A \in \mathbb{C}^{m\times n}$ be a matrix.*

1. *If $V\Lambda V^{\mathsf{H}}$ is a compact eigenvalue decomposition of $A^{\mathsf{H}}A$ and $U = AV\Lambda^{-\frac{1}{2}}$, then $U\Lambda^{\frac{1}{2}}V^{\mathsf{H}}$ is a compact SVD of A.*

2. *If $U\Lambda U^{\mathsf{H}}$ is a compact eigenvalue decomposition of AA^{H} and $V = A^{\mathsf{H}}U\Lambda^{-\frac{1}{2}}$, then $U\Lambda^{\frac{1}{2}}V^{\mathsf{H}}$ is a compact SVD of A.*

Proof. We only prove 1. By assumption, $V^{\mathsf{H}}V = I$, $A^{\mathsf{H}}A = V\Lambda V^{\mathsf{H}}$, and $U = AV\Lambda^{-\frac{1}{2}}$. Hence

$$U^{\mathsf{H}}U \ =\ (AV\Lambda^{-\frac{1}{2}})^{\mathsf{H}}(AV\Lambda^{-\frac{1}{2}}) \ =\ \Lambda^{-\frac{1}{2}}(V^{\mathsf{H}}A^{\mathsf{H}}AV)\Lambda^{-\frac{1}{2}} \ =\ \Lambda^{-\frac{1}{2}}\Lambda\Lambda^{-\frac{1}{2}} \ =\ I. \tag{2.5}$$

Thus

$$U^{\mathsf{H}}AV \ =\ U^{\mathsf{H}}(U\Lambda^{\frac{1}{2}}) \ =\ \Lambda^{\frac{1}{2}}.$$

Meanwhile, $\|U\| = 1$ by (2.5), $\|V\| = 1$ because $V^{\mathsf{H}}V = I$, and

$$\|A\|_{\mathrm{F}}^2 \ =\ \mathrm{tr}(A^{\mathsf{H}}A) \ =\ \mathrm{tr}(\Lambda) \ =\ \|\Lambda^{\frac{1}{2}}\|_{\mathrm{F}}^2.$$

Therefore, Lemma 5 ensures

$$A \ =\ U\Lambda^{\frac{1}{2}}V^{\mathsf{H}}.$$

Hence $U\Lambda^{\frac{1}{2}}V^{\mathsf{H}}$ is a compact SVD of A. □

Remark 8. *The major point of the proof is to show that $U\Lambda^{\frac{1}{2}}V^{\mathsf{H}} = A$. Here we use Lemma 5, but there are other ways to prove it.*

Corollary 1. *Let $A \in \mathbb{C}^{m\times n}$ be a matrix.*

1. *If $V\Lambda V^{\mathsf{H}}$ is an eigenvalue decomposition of $A^{\mathsf{H}}A$ such that the diagonal entries of Λ are descending. The there exist $U \in \mathbb{C}^{m\times m}$ and $\Sigma \in \mathbb{R}^{m\times n}$ such that $U\Sigma V^{\mathsf{H}}$ is an SVD of A.*

2. *If $U\Lambda U^{\mathsf{H}}$ is an eigenvalue decomposition of AA^{H} such that the diagonal entries of Λ are descending. The there exist $V \in \mathbb{C}^{n\times n}$ and $\Sigma \in \mathbb{R}^{m\times n}$ such that $U\Sigma V^{\mathsf{H}}$ is an SVD of A.*

Proof. We only prove 1. Suppose that $\operatorname{rank}(A) = r$. Let $\hat{\Lambda} = \operatorname{diag}(\Lambda_{1,1}, \ldots, \Lambda_{r,r})$ and $\hat{V}$ be the first r columns of V. Then $\hat{V}\hat{\Lambda}\hat{V}^{\mathsf{H}}$ is a compact eigenvalue decomposition of $A^{\mathsf{H}}A$. With $\hat{U} = A\hat{V}\hat{\Lambda}^{-\frac{1}{2}}$, we know from Theorem 5 that $\hat{U}\hat{\Lambda}^{\frac{1}{2}}\hat{V}^{\mathsf{H}}$ is a compact SVD of A. Let $\tilde{U} \in \mathbb{C}^{m\times(m-r)}$ be any matrix such that $(\hat{U}\ \tilde{U})$ is unitary. Then

$$(\hat{U}\ \tilde{U}) \begin{pmatrix} \hat{\Lambda}^{\frac{1}{2}} & 0 \\ 0 & 0 \end{pmatrix} V^{\mathsf{H}}$$

is an SVD of A. □

2.2.3 The Wielandt-Lanczos approach [8]

Lemma 6. *Given a matrix $A \in \mathbb{C}^{m\times n}$, define its Jordan-Wielandt form [9] to be*

$$J = \begin{pmatrix} 0 & A \\ A^{\mathsf{H}} & 0 \end{pmatrix}. \tag{2.6}$$

Then the characteristic polynomial of J is

$$p(\sigma) \;=\; \sigma^{m-n}\det(\sigma^2 I_n - A^{\mathsf{H}}A) \;=\; \sigma^{n-m}\det(\sigma^2 I_m - AA^{\mathsf{H}}). \tag{2.7}$$

If the nonzero eigenvalues of AA^{H} (i.e., those of $A^{\mathsf{H}}A$) are $\lambda_1, \ldots, \lambda_r$, multiplicity included, then the nonzero eigenvalues of J are $\sqrt{\lambda_1}, -\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_r}, -\sqrt{\lambda_r}$, multiplicity included.

Proof. We only prove the first equality in (2.7). For any $\sigma \neq 0$,

$$\begin{pmatrix} I_m & 0 \\ \sigma^{-1}A^{\mathsf{H}} & I_n \end{pmatrix} \begin{pmatrix} \sigma I_m & -A \\ -A^{\mathsf{H}} & \sigma I_n \end{pmatrix} = \begin{pmatrix} \sigma I_m & -A \\ 0 & \sigma I_n - \sigma^{-1}A^{\mathsf{H}}A \end{pmatrix}.$$

Taking determinants, we have

$$\det(\sigma I - J) \;=\; \det(\sigma I_m)\det(\sigma I_n - \sigma^{-1}A^{\mathsf{H}}A) \;=\; \sigma^{m-n}\det(\sigma^2 I_n - A^{\mathsf{H}}A). \tag{2.8}$$

In (2.8), two rational functions are equal for all $\sigma \neq 0$. Hence they are indeed identical. □

Theorem 6. *Consider matrices $A \in \mathbb{C}^{m\times n}$, $\Sigma \in \mathbb{R}^{r\times r}$, $U_i \in \mathbb{C}^{m\times r}$, and $V_i \in \mathbb{C}^{n\times r}$ $(i = 1, 2)$. Suppose that Σ is a diagonal matrix whose diagonal entries are positive. Then*

$$\begin{pmatrix} U_1 & U_2 \\ V_1 & V_2 \end{pmatrix} \begin{pmatrix} \Sigma & 0 \\ 0 & -\Sigma \end{pmatrix} \begin{pmatrix} U_1 & U_2 \\ V_1 & V_2 \end{pmatrix}^{\mathsf{H}} \tag{2.9}$$

is a compact eigenvalue decomposition of the Jordan-Wielandt matrix J in (2.6) if and only if both $(\sqrt{2}U_1)\Sigma(\sqrt{2}V_1)^{\mathsf{H}}$ and $(-\sqrt{2}U_2)\Sigma(\sqrt{2}V_2)^{\mathsf{H}}$ are compact SVDs of A.

Proof. 1. Assume that (2.9) is a compact eigenvalue decomposition of J. We will prove that $(\sqrt{2}U_1)\Sigma(\sqrt{2}V_1)^{\mathsf{H}}$ is a compact SVD of A, and the other one can be discussed similarly. It suffices to show that

$$U_1^{\mathsf{H}}U_1 \;=\; V_1^{\mathsf{H}}V_1 \;=\; \frac{I_r}{2}, \quad U_1\Sigma V_1^{\mathsf{H}} \;=\; \frac{A}{2}. \tag{2.10}$$

Due to the compact eigenvalue decomposition (2.9) of J, the columns of $[U_1; V_1]$ are eigenvectors of J associated with all its r positive eigenvalues,[1] and

$$J \begin{pmatrix} U_1 \\ V_1 \end{pmatrix} = \begin{pmatrix} U_1 \\ V_1 \end{pmatrix} \Sigma, \tag{2.11}$$

This implies $AV_1 = U_1\Sigma$ and $A^{\mathsf{H}}U_1 = \Sigma V_1$, which can be reformulated as

$$J \begin{pmatrix} U_1 \\ -V_1 \end{pmatrix} = \begin{pmatrix} U_1 \\ -V_1 \end{pmatrix} (-\Sigma), \tag{2.12}$$

i.e., the columns of $[U_1; -V_1]$ are eigenvectors of J associated with the negative eigenvalues. Hence the columns of $[U_1; V_1]$ and those of $[U_1; -V_1]$ are orthogonal. Thus

$$U_1^{\mathsf{H}}U_1 - V_1^{\mathsf{H}}V_1 \;=\; (U_1^{\mathsf{H}} \;\; V_1^{\mathsf{H}}) \begin{pmatrix} U_1 \\ -V_1 \end{pmatrix} \;=\; 0.$$

With (2.9) being a compact eigenvalue decomposition, we also have $U_1^{\mathsf{H}}U_1 + V_1^{\mathsf{H}}V_1 = I_r$. Hence $U_1^{\mathsf{H}}U_1 = V_1^{\mathsf{H}}V_1 = I_r/2$, which is the first equality in (2.10). To establish the second one, define

$$\bar{U} \;=\; \begin{pmatrix} U_1 & U_1 \\ V_1 & -V_1 \end{pmatrix}, \quad \bar{\Sigma} \;=\; \begin{pmatrix} \Sigma & 0 \\ 0 & -\Sigma \end{pmatrix}.$$

Then

$$\bar{U}^{\mathsf{H}}\bar{U} \;=\; \begin{pmatrix} U_1^{\mathsf{H}}U_1 + V_1^{\mathsf{H}}V_1 & U_1^{\mathsf{H}}U_1 - V_1^{\mathsf{H}}V_1 \\ U_1^{\mathsf{H}}U_1 - V_1^{\mathsf{H}}V_1 & U_1^{\mathsf{H}}U_1 + V_1^{\mathsf{H}}V_1 \end{pmatrix} \;=\; \begin{pmatrix} I_r & 0 \\ 0 & I_r \end{pmatrix}. \tag{2.13}$$

Meanwhile, we can reformulate (2.11)–(2.12) as $J\bar{U} = \bar{U}\bar{\Sigma}$. Therefore,

$$\bar{U}^{\mathsf{H}}J\bar{U} \;=\; \bar{\Sigma}.$$

Note that $\|J\|_{\mathrm{F}} = \|\bar{\Sigma}\|_{\mathrm{F}}$ according to the compact eigenvalue decomposition (2.9) and $\|\bar{U}\| = 1$ due to (2.13). Thus Lemma 5 renders

$$J \;=\; \bar{U}\bar{\Sigma}\bar{U}^{\mathsf{H}}, \tag{2.14}$$

from which we can obtain $A = 2U_1\Sigma V_1^{\mathsf{H}}$ by straightforward calculations.

2. Assume that both $(\sqrt{2}U_1)\Sigma(\sqrt{2}V_1)^{\mathsf{H}}$ and $(-\sqrt{2}U_2)\Sigma(\sqrt{2}V_2)^{\mathsf{H}}$ are compact SVDs of A. Then we have (2.10) and

$$U_2^{\mathsf{H}}U_2 \;=\; V_2^{\mathsf{H}}V_2 \;=\; \frac{I_r}{2}, \quad -U_2\Sigma V_2^{\mathsf{H}} \;=\; \frac{A}{2}. \tag{2.15}$$

To prove that (2.9) is a compact SVD for J, it suffices to show

$$\begin{pmatrix} 0 & A \\ A^{\mathsf{H}} & 0 \end{pmatrix} = \begin{pmatrix} U_1 & U_2 \\ V_1 & V_2 \end{pmatrix} \begin{pmatrix} \Sigma & 0 \\ 0 & -\Sigma \end{pmatrix} \begin{pmatrix} U_1 & U_2 \\ V_1 & V_2 \end{pmatrix}^{\mathsf{H}}, \quad \begin{pmatrix} U_1 & U_2 \\ V_1 & V_2 \end{pmatrix}^{\mathsf{H}} \begin{pmatrix} U_1 & U_2 \\ V_1 & V_2 \end{pmatrix} = \begin{pmatrix} I_r & 0 \\ 0 & I_r \end{pmatrix},$$

which resolve to

$$\begin{cases} U_1\Sigma U_1^{\mathsf{H}} - U_2\Sigma U_2^{\mathsf{H}} \;=\; 0, \quad V_1\Sigma V_1^{\mathsf{H}} - V_2\Sigma V_2^{\mathsf{H}} \;=\; 0, \quad U_1\Sigma V_1^{\mathsf{H}} - U_2\Sigma V_2^{\mathsf{H}} \;=\; A, & (2.16) \\ U_1^{\mathsf{H}}U_1 + V_1^{\mathsf{H}}V_1 \;=\; I_r, \quad U_2^{\mathsf{H}}U_2 + V_2^{\mathsf{H}}V_2 \;=\; I_r, \quad U_1^{\mathsf{H}}U_2 + V_1^{\mathsf{H}}V_2 \;=\; 0. & (2.17) \end{cases}$$

[1] Recall that the MATLAB-style notation $[a; b]$ denotes a vertical array with a and b being its entries.

Since $U_1\Sigma V_1^{\mathsf{H}} = -U_2\Sigma V_2^{\mathsf{H}} = A/2$, Theorem 3 implies the existence of a unitary matrix $W \in \mathbb{C}^{r\times r}$ such that

$$U_2 \;=\; -U_1W, \quad V_2 \;=\; V_1W, \quad \Sigma W \;=\; W\Sigma.$$

Hence

$$U_2\Sigma U_2^{\mathsf{H}} \;=\; (-U_1W)\Sigma(-U_1W)^{\mathsf{H}} \;=\; U_1W\Sigma W^{\mathsf{H}}U_1^{\mathsf{H}} \;=\; U_1\Sigma WW^{\mathsf{H}}U_1^{\mathsf{H}} \;=\; U_1\Sigma U_1^{\mathsf{H}},$$

which implies $U_1\Sigma U_1^{\mathsf{H}} - U_2\Sigma U_2^{\mathsf{H}} = 0$. Similarly, $V_1\Sigma V_1^{\mathsf{H}} - V_2\Sigma V_2^{\mathsf{H}} = 0$. In addition,

$$U_1^{\mathsf{H}}U_2 + V_1^{\mathsf{H}}V_2 \;=\; -U_1^{\mathsf{H}}U_1W + V_1^{\mathsf{H}}V_1W \;=\; 0,$$

where we use the fact that $U_1^{\mathsf{H}}U_1 = V_1^{\mathsf{H}}V_1 = I_r/2$ from (2.10). By (2.10) and (2.15), we also have

$$U_1^{\mathsf{H}}U_1 + V_1^{\mathsf{H}}V_1 \;=\; U_2^{\mathsf{H}}U_2 + V_2^{\mathsf{H}}V_2 \;=\; I_r, \quad U_1\Sigma V_1^{\mathsf{H}} - U_2\Sigma V_2^{\mathsf{H}} \;=\; A.$$

All the equalities in (2.16)–(2.17) have been justified. The proof is complete. □

Remark 9. *According to* (2.13) *and* (2.14), $\bar{U}\bar{\Sigma}\bar{U}^{\mathsf{H}}$ *is indeed a compact eigenvalue decomposition of* J.

3 Decompose a matrix into partial isometries [7]

Definition 3. A matrix $A \in \mathbb{C}^{m\times n}$ is said to be a partial isometry if $\|Ax\| = \|x\|$ for each $x \in \operatorname{range}(A^{\mathsf{H}})$ (i.e., $x \in (\ker A)^{\perp}$).

The following proposition collects various characterizations of partial isometries.

Proposition 1. *For any* $A \in \mathbb{C}^{m\times n}$, *the following statements are equivalent.*

1. A *is a partial isometry.*
2. A^{H} *is a partial isometry.*
3. $A^{\mathsf{H}}A$ *is an orthogonal projection.*
4. AA^{H} *is an orthogonal projection.*
5. $A^{\mathsf{H}}AA^{\mathsf{H}} = A^{\mathsf{H}}$.
6. $AA^{\mathsf{H}}A = A$.
7. *All the nonzero singular values of* A *are* 1.
8. *The linear operator* $x \mapsto Ax$ *is an isometric isomorphism from* $\operatorname{range}(A^{\mathsf{H}})$ *to* $\operatorname{range}(A)$.
9. *The linear operator* $x \mapsto A^{\mathsf{H}}x$ *is an isometric isomorphism from* $\operatorname{range}(A)$ *to* $\operatorname{range}(A^{\mathsf{H}})$.

Proof. 1 $\Rightarrow$ 8. Obvious.

8 $\Rightarrow$ 9. Take any $x \in \mathrm{range}(A)$. There is a $y \in \mathrm{range}(A^{\mathsf{H}})$ such that $x = Ay$. By assumption, $\|y\| = \|x\|$. Hence

$$\|A^{\mathsf{H}}x\| \;\geq\; (y/\|y\|)^{\mathsf{H}}A^{\mathsf{H}}x \;=\; \frac{1}{\|x\|}y^{\mathsf{H}}A^{\mathsf{H}}x \;=\; \frac{1}{\|x\|}x^{\mathsf{H}}x \;=\; \|x\|.$$

For any unit vector $z \in \mathbb{C}^n$, let z' be its orthogonal projection on $\mathrm{range}(A^{\mathsf{H}})$. Then

$$\|Az\| \;=\; \|Az'\| \;=\; \|z'\| \;\leq\; \|z\| \;=\; 1.$$

Thus

$$\|A^{\mathsf{H}}x\| \;=\; \max_{\|z\|=1} z^{\mathsf{H}}A^{\mathsf{H}}x \;\leq\; \max_{\|z\|=1} \|Az\|\|x\| \;\leq\; \|x\|.$$

9 $\Rightarrow$ 2. Obvious.

2 $\Rightarrow$ 3. Since $A^{\mathsf{H}}A$ is Hermitian, it suffices to show that it is idempotent. We only need to prove that $x^{\mathsf{H}}(A^{\mathsf{H}}A)^2y = x^{\mathsf{H}}A^{\mathsf{H}}Ay$ for any x and $y \in \mathbb{R}^n$, or equivalently, $u^{\mathsf{H}}AA^{\mathsf{H}}v = u^{\mathsf{H}}v$ for any u and $v \in \mathrm{range}(A)$. By assumption, For any u, $v \in \mathrm{range}(A)$, we have $\|A^{\mathsf{H}}u\| = \|u\|$, $\|A^{\mathsf{H}}v\| = \|v\|$, and $\|A^{\mathsf{H}}(u+v)\| = \|u+v\|$. Squaring the last equality, we obtain $u^{\mathsf{H}}AA^{\mathsf{H}}v = u^{\mathsf{H}}v$.

3 $\Rightarrow$ 5. Since $A^{\mathsf{H}}A$ is a projection, we have $A^{\mathsf{H}}Ax = x$ for all $x \in \mathrm{range}(A^{\mathsf{H}}A) = \mathrm{range}(A^{\mathsf{H}})$. Hence $A^{\mathsf{H}}AA^{\mathsf{H}} = A^{\mathsf{H}}$.

5 $\Rightarrow$ 4. AA^{H} is Hermitian, and $(AA^{\mathsf{H}})^2 = AA^{\mathsf{H}}AA^{\mathsf{H}} = AA^{\mathsf{H}}$.

4 $\Rightarrow$ 6. Similar to 3 $\Rightarrow$ 5.

6 $\Rightarrow$ 7. Since AA^{H} is positive semidefinite and $(AA^{\mathsf{H}})^2 = AA^{\mathsf{H}}AA^{\mathsf{H}} = AA^{\mathsf{H}}$, we know that all the nonzero eigenvalues of AA^{H} are 1.

7 $\Rightarrow$ 1. Let $r = \mathrm{rank}(A)$. If $r = 0$, then the conclusion is trivially true. Otherwise, A has a compact SVD of the form UV^{H}, where $U \in \mathbb{C}^{m\times r}$ and $V \in \mathbb{C}^{n\times r}$ satisfy $U^{\mathsf{H}}U = V^{\mathsf{H}}V = I_r$. Note that the columns of V is an orthonormal basis of $\mathrm{range}(A^{\mathsf{H}})$. Therefore, for any $x \in \mathrm{range}(A^{\mathsf{H}})$, $\|Ax\| = \|UV^{\mathsf{H}}x\| = \|V^{\mathsf{H}}x\| = \|x\|$. □

Theorem 7. *For any Hermitian matrix $A \in \mathbb{C}^{n\times n}$, there exists a unique decomposition*

$$A \;=\; \sum_{i=1}^{k} \lambda_i P_i \tag{3.1}$$

such that

1. *$\{\lambda_i\}$ are all real numbers and $\lambda_1 > \cdots > \lambda_k$.*
2. *$\{P_i\}$ are all orthogonal projections, $P_iP_j = 0$ for any distinct i and j, and $\sum_{i=1}^{k} P_i = I$.*

Proof. The existence is easy to establish by any eigenvalue decomposition of A. We only prove the uniqueness.

Consider any decomposition in the form of (3.1). For each i, let V_i be a matrix whose columns form an orthonormal basis of $\mathrm{range}(P_i)$. Then $V_i^{\mathsf{H}}V_i$ is an identity matrix, and $P_i = V_iV_i^{\mathsf{H}}$. For any distinct i and j,

$$V_i^{\mathsf{H}}V_j \;=\; (V_i^{\mathsf{H}}V_i)V_i^{\mathsf{H}}V_j(V_j^{\mathsf{H}}V_j) \;=\; V_i^{\mathsf{H}}P_iP_jV_j \;=\; 0.$$

Define

$$V = (V_1 \ \cdots \ V_k).$$

Then the columns of V are orthonormal. In addition,

$$VV^{\mathsf{H}} \ = \ \sum_{i=1}^{k} V_i V_i^{\mathsf{H}} \ = \ \sum_{i=1}^{k} P_i \ = \ I.$$

Thus V is a unitary matrix. In addition,

$$A \ = \ \sum_{i=1}^{k} \lambda_i P_i \ = \ \sum_{i=1}^{k} \lambda_i V_i V_i^{\mathsf{H}} \ = \ \sum_{i=1}^{k} V_i \Lambda_i V_i^{\mathsf{H}} \ = \ V \Lambda V^{\mathsf{H}},$$

where $\Lambda_i = \lambda_i V_i^{\mathsf{H}} V_i$, and Λ is the block diagonal matrix whose diagonal blocks are Λ_i. Note that Λ indeed a diagonal matrix since each Λ_i is diagonal. Thus $V \Lambda V^{\mathsf{H}}$ is an eigenvalue decomposition of A, with $\lambda_1, \ldots, \lambda_k$ being all the distinct eigenvalues, ranked in the descending order. Moreover, for each i, the columns of V_i form an orthonormal basis of the eigenspace associated with λ_i, and hence P_i is the orthogonal projection onto this eigenspace. In this way, $\{\lambda_i\}$ and $\{P_i\}$ are uniquely determined by A. □

Theorem 8. *For any nonzero matrix $A \in \mathbb{C}^{m \times n}$, there exists a unique decomposition*

$$A \ = \ \sum_{i=1}^{k} \sigma_i A_i \tag{3.2}$$

such that

1. $\sigma_1 > \cdots > \sigma_k > 0$;
2. $\{A_i\}$ *are all partial isometries, with* $A_i A_j^{\mathsf{H}}$ *and* $A_i^{\mathsf{H}} A_j$ *both being zero for any distinct* i *and* j.

Proof. The existence is easy to establish by an SVD of A. We only prove the uniqueness.

Consider any decomposition in the form of (3.2). Since $A_i^{\mathsf{H}} A_j = 0$ for any distinct i and j, we have

$$A^{\mathsf{H}} A \ = \ \left(\sum_{i=1}^{k} \sigma_i A_i \right)^{\mathsf{H}} \left(\sum_{i=1}^{k} \sigma_i A_i \right) \ = \ \sum_{i=1}^{k} \sigma_i^2 A_i^{\mathsf{H}} A_i. \tag{3.3}$$

For each i, $A_i^{\mathsf{H}} A_i$ is an orthogonal projection as A_i is a partial isometry (see 3 of Proposition 1). Hence (3.3) is a decomposition specified in Theorem 7. Due to the uniqueness part of Theorem 7, $\sigma_1, \ldots, \sigma_k$ and $A_1^{\mathsf{H}} A_1, \ldots, A_k^{\mathsf{H}} A_k$ are uniquely determined by A.

Now consider any two decompositions in the form of (3.2). According to what is proved above, we can formulate the decompositions as

$$A \ = \ \sum_{i=1}^{k} \sigma_i A_i \quad \text{and} \quad A \ = \ \sum_{i=1}^{k} \sigma_i \tilde{A}_i, \tag{3.4}$$

and we have $A_i^{\mathsf{H}} A_i = \tilde{A}_i^{\mathsf{H}} \tilde{A}_i$ for each i. For any distinct i and j,

$$A_i \tilde{A}_j^{\mathsf{H}} \ = \ (A_i A_i^{\mathsf{H}} A_i) \tilde{A}_j^{\mathsf{H}} \ = \ (A_i \tilde{A}_i^{\mathsf{H}} \tilde{A}_i) \tilde{A}_j^{\mathsf{H}} \ = \ 0,$$

where the first equality is because A_i is a partial isometry (see 6 of Proposition 1), and the second is because $\tilde{A}_i\tilde{A}_j^{\mathsf{H}} = 0$. Similarly, $\tilde{A}_i A_j^{\mathsf{H}} = 0$. Hence

$$(A_i - \tilde{A}_i)(A_j - \tilde{A}_j)^{\mathsf{H}} \;=\; 0.$$

Thus

$$\left[\sum_{i=1}^{k} \sigma_i(A_i - \tilde{A}_i)\right]\left[\sum_{i=1}^{k} \sigma_i(A_i - \tilde{A}_i)\right]^{\mathsf{H}} \;=\; \sum_{i=1}^{k} \sigma_i^2 (A_i - \tilde{A}_i)(A_i - \tilde{A}_i)^{\mathsf{H}}. \tag{3.5}$$

According to (3.4), the left-hand side of (3.5) is zero. Hence $A_i = \tilde{A}_i$ for each i. Therefore, the two decompositions in (3.4) are identical. The proof is complete. □

Theorem 8 is indeed the matrix version of the following theorem.

Theorem 9 ([7, Theorem 1])**.** *Suppose that X and Y are finite dimensional Hilbert spaces, and $T : X \to Y$ is a linear operator. Then there exist unique orthogonal decompositions*

$$\operatorname{range}(T^*) = X_1 \oplus \cdots \oplus X_k, \quad \operatorname{range}(T) = Y_1 \oplus \cdots \oplus Y_k,$$

scalars $\sigma_1 > \cdots > \sigma_k$, and isometries $T_i : X_i \to Y_i$ $(i = 1, \ldots, k)$ such that

$$T|_{X_i} \;=\; \sigma_i T_i \quad \textit{for each} \quad i \in \{1, \ldots, k\}.$$

4 SVD as a change of basis

Theorem 10 collects basics facts about the matrix representation of linear operators between finite dimensional vector spaces on $\mathbb{C}$.

Theorem 10. *Suppose that X is finite dimensional vector space on $\mathbb{C}$ with $\{x_1, \ldots, x_n\}$ being its basis and $C_X : X \to \mathbb{C}^n$ being the map from any point in X to its coordinate under this basis; Y, $\{y_1, \ldots, y_m\}$, and $C_Y : Y \to \mathbb{C}^m$ are similar. Consider a linear operator $T : X \to Y$.*

1. *There is a unique matrix $A \in \mathbb{C}^{m\times n}$ that represents T under the aforementioned bases of X and Y in the sense that*

   $$AC_X(x) = C_Y(T(x)) \quad \textit{for all} \quad x \in X,$$

   meaning that applying T to x is equivalent to multiplying its coordinate by A. Indeed, the i-th column of A is $A_i = C_Y(T(x_i))$, namely the coordinate of $T(x_i)$.

2. *A has full row rank if and only if T is injective; A has full column rank if and only if T is surjective; when $m = n$, T is invertible if and only if A is invertible, and A^{-1} represents $T^{-1} : Y \to X$ under the aforementioned bases for X and Y.*

3. *If the bases $\{x_1, \ldots, x_n\}$ and $\{y_1, \ldots, y_m\}$ are both orthonormal, then A^{H} represents $T^* : Y \to X$ under these bases. (Here, as usual, T^* denotes the Hilbert adjoint of T, which is the unique linear operator from Y to X such that $\langle y, T(x)\rangle_Y = \langle T^*(y), x\rangle_X$.)*

4. *Let $\{x'_1, \ldots, x'_n\}$ be a basis for X, $\{y'_1, \ldots, y'_m\}$ be a basis for Y, and $B \in \mathbb{C}^{m\times n}$ be the matrix representation of T under such bases. Then $A = Q^{-1}BP$, where $P \in \mathbb{C}^{n\times n}$ and $Q \in \mathbb{C}^{m\times m}$ are the changing of basis matrices such that*

$$(x_1 \ \cdots \ x_n) = (x'_1 \ \cdots \ x'_n)P, \qquad (y_1 \ \cdots \ y_m) = (y'_1 \ \cdots \ y'_m)Q.$$

With the view point presented in Theorem 10, we can understand SVD as follows.

Let $A \in \mathbb{C}^{m\times n}$ be a nonzero matrix, and $U\Sigma V^{\mathsf{H}}$ be its SVD. Consider the linear operator $T : x \mapsto Ax$ from $\mathbb{C}^n$ to $\mathbb{C}^m$. Then A represents T under the canonical bases. The aforementioned SVD suggests a change of basis that can simplify the matrix representation of T: if we take the columns of V as the basis for $\mathbb{C}^n$ and those of U as the basis for $\mathbb{C}^m$, then the matrix representation of T under these bases is Σ, which is diagonal.

Recall the decompositions

$$\mathbb{C}^n \ = \ \ker(A) \oplus \operatorname{range}(A^{\mathsf{H}}), \quad \mathbb{C}^m \ = \ \ker(A^{\mathsf{H}}) \oplus \operatorname{range}(A).$$

When T acts on $\mathbb{C}^n$, it drops out the information in $\ker(A)$, and provides no information in $\ker(A^{\mathsf{H}})$. Consequently, T is not an isomorphism if either $\ker(A)$ or $\ker(A^{\mathsf{H}})$ is nonzero. The restriction $\hat{T} : \operatorname{range}(A^{\mathsf{H}}) \to \operatorname{range}(A)$ with $\hat{T}(x) = Ax$ is however always an isomorphism. Suppose that $\hat{U}\hat{\Sigma}\hat{V}^{\mathsf{H}}$ is a compact SVD of A. Then the columns of $\hat{U}$ form an orthogonal basis for $\operatorname{range}(A)$, and those of $\hat{V}$ form an orthonormal basis for $\operatorname{range}(A^{\mathsf{H}})$. Under these bases, $\hat{T}$ is represented by $\hat{\Sigma}$. The matrix representation for its inverse $\hat{T}^{-1} : \operatorname{range}(A) \to \operatorname{range}(A^{\mathsf{H}})$ is $\hat{\Sigma}^{-1}$.

The operator $T^+ : \mathbb{C}^m \to \mathbb{C}^n$ defined by

$$T^+|_{\operatorname{range}(A)} = \hat{T}^{-1}, \quad T^+|_{\ker(A^{\mathsf{H}})} = 0$$

is called the Moore-Penrose pseudoinverse of T. The matrix representation of T^+ under the canonical bases is called the Moore-Penrose pseudoinverse of A, which turns out to be $\hat{U}\hat{\Sigma}^{-1}\hat{V}^{\mathsf{H}}$.

Given any nonzero linear operator T between two finite dimensional Hilbert spaces, the restriction of T on $\ker(T)^{\perp} = \operatorname{range}(T^*)$ is a bijection between $\operatorname{range}(T^*)$ and $\operatorname{range}(T)$. SVD tells us that such a restriction can be represented by a positive diagonal matrix under properly chosen orthonormal bases for $\operatorname{range}(T^*)$ and $\operatorname{range}(T)$. This can also lead us to Theorem 9.

5 Examples of applications

Proposition 2 (Polar decomposition)**.** *Let $A \in \mathbb{C}^{m\times n}$ be a matrix.*

1. *If $m \geq n$, there exists a positive semidefinite matrix $P \in \mathbb{C}^{n\times n}$ and a matrix $U \in \mathbb{C}^{m\times n}$ such that $A = UP$ and $U^{\mathsf{H}}U = I_n$; there also exists a positive semidefinite matrix $Q \in \mathbb{C}^{m\times m}$ and a matrix $V \in \mathbb{C}^{m\times n}$ such that $A = QV$ and $V^{\mathsf{H}}V = I_n$. In this case, $P = (A^{\mathsf{H}}A)^{\frac{1}{2}}$.*

2. *If $n \geq m$, there exists a positive semidefinite matrix $P \in \mathbb{C}^{n\times n}$ and a matrix $U \in \mathbb{C}^{m\times n}$ such that $A = UP$ and $UU^{\mathsf{H}} = I_m$; there also exists a positive semidefinite matrix $Q \in \mathbb{C}^{m\times m}$ and a matrix $V \in \mathbb{C}^{m\times n}$ such that $A = QV$ and $VV^{\mathsf{H}} = I_m$. In this case, $Q = (AA^{\mathsf{H}})^{\frac{1}{2}}$.*

If A is real, we can require P, U, Q, and V to be real.

Proof. We only prove 1. Let $W\Sigma Z^{\mathsf{H}}$ be an SVD of A.

Note that the last $m-n$ rows of Σ are zero. Let $\hat{\Sigma}$ be the first n rows of Σ, and $\hat{W}$ be the first n columns of W. Then $A = \hat{W}\hat{\Sigma}Z^{\mathsf{H}}$. Define $U = \hat{W}Z^{\mathsf{H}}$ and $P = Z\hat{\Sigma}Z^{\mathsf{H}}$. Then P is positive semidefinite, and

$$A \;=\; \hat{W}\hat{\Sigma}Z \;=\; UP, \quad U^{\mathsf{H}}U \;=\; Z\hat{W}^{\mathsf{H}}\hat{W}Z^{\mathsf{H}} \;=\; ZZ^{\mathsf{H}} \;=\; I_n.$$

Consequently, $A^{\mathsf{H}}A = P^{\mathsf{H}}U^{\mathsf{H}}UP = P^2$, and hence $P = (A^{\mathsf{H}}A)^{\frac{1}{2}}$.

Let $\bar{\Sigma} = (\Sigma \;\; 0_{m\times(m-n)})$ and $\bar{Z} = (Z \;\; 0_{n\times(m-n)})$. Then $A = W\bar{\Sigma}\bar{Z}^{\mathsf{H}}$. Define $Q = W\bar{\Sigma}W^{\mathsf{H}}$ and $V = W\bar{Z}^{\mathsf{H}}$. Then Q is positive semidefinite, and

$$A \;=\; W\bar{\Sigma}\bar{Z}^{\mathsf{H}} \;=\; QV, \quad V^{\mathsf{H}}V \;=\; \bar{Z}W^{\mathsf{H}}W\bar{Z}^{\mathsf{H}} \;=\; \bar{Z}\bar{Z}^{\mathsf{H}} \;=\; ZZ^{\mathsf{H}} \;=\; I_n.$$

If A is real, then W, Σ, and Z can all be real, ensuring P, U, Q, and V to be real. □

Proposition 3 ([3])**.** *Let H be the Hermitian part of a matrix $A \in \mathbb{C}^{n\times n}$. Enumerating the eigenvalues of H as $\lambda_1(H) \geq \cdots \geq \lambda_n(H)$, and the singular values of A as $\sigma_i(A) \geq \cdots \geq \sigma_n(A)$, we have $\sigma_i(A) \geq \lambda_i(H)$ for each $i = 1, \ldots, n$.*

Proof. By Theorem 2, there exists a positive semidefinite matrix $P \in \mathbb{C}^{n\times n}$ and a unitary matrix $U \in \mathbb{C}^{n\times n}$ such that $A = UP$. For any unit vector $x \in \mathbb{C}^n$,

$$x^{\mathsf{H}}Hx = \frac{1}{2}x^{\mathsf{H}}(A^{\mathsf{H}} + A)x = \Re(x^{\mathsf{H}}Ax) \leq |x^{\mathsf{H}}UPx| \leq \|Px\| = (x^{\mathsf{H}}P^2x)^{\frac{1}{2}} = (x^{\mathsf{H}}A^{\mathsf{H}}Ax)^{\frac{1}{2}}.$$

Therefore, by the Courant-Fischer-Weyl min-max principle, we know that

$$\lambda_i(H) \;\leq\; \lambda_i(A^{\mathsf{H}}A)^{\frac{1}{2}} \;=\; \sigma_i(A).$$

□

Proposition 4. *For matrices A_1 and $A_2 \in \mathbb{C}^{m\times n}$, $A_1^{\mathsf{H}}A_1 = A_2^{\mathsf{H}}A_2$ if and only if there exists a unitary matrix $U \in \mathbb{C}^{m\times m}$ such that $A_2 = UA_1$.*

Proof. The "if" part is trivial. We focus on the "only if" part. Let $V\Lambda V^{\mathsf{H}}$ be an eigenvalue decomposition of $A_1^{\mathsf{H}}A_1 = A_2^{\mathsf{H}}A_2$ such that the diagonal entries of Λ are descending. By Corollary 1, there exists W_1, $W_2 \in \mathbb{C}^{m\times m}$ and $\Sigma \in \mathbb{R}^{m\times n}$ such that $A_1 = W_1\Sigma V^{\mathsf{H}}$ and $A_2 = W_2\Sigma V^{\mathsf{H}}$. Set $U = W_2W_1^{\mathsf{H}}$. □

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